

Exploring Effective Real-Time Ergonomic Guidance Methods for Immersive Virtual Reality Workspaces

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ABSTRACT

Studying and working in Virtual Reality (VR) provides an immersive experience and enables free body movement for input. However, poor working posture in VR can result in discomfort and even lead to musculoskeletal problems over time. In this work, we explore how to use Mediapipe BlazePose to provide effective real-time ergonomic posture guidance methods in a virtual office environment using (1) auditory, (2) visual, and (3) combined auditory-visual cues. We do this in a within-subject design user study comparing these three conditions and a baseline condition where no guidance is provided. Our results indicate that all three guidance conditions significantly improved users' ergonomic posture maintenance compared with the baseline condition. However, we found that the combined auditory-visual guidance is the most effective and preferred method. We also discuss multi-modal interaction methods, privacy concerns, and directions for future research.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; **Graphical user interfaces**; **Empirical studies in HCI**.

KEYWORDS

Ergonomic guidance, Virtual Reality, Immersive Workspace

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1 INTRODUCTION

Virtual Reality (VR) workspaces can provide better immersion than traditional office environments [5] and are considered as an option to replace PCs in the physical workspace [13]. However, ergonomic posture-related issues exist in VR workspaces, just as in the physical workspace. For example, prolonged exposure can lead to musculoskeletal discomfort [13] and various pains, so more research needs to be done on real-time comfort assessment [7]. These issues could be avoided if poor posture can be detected and feedback given to the user to allow them to adjust their ergonomic posture state.

In this paper, we present an augmented guiding system that can provide real-time posture adjustment feedback in a VR work environment based on the Rapid Upper Limb Assessment (RULA) [10] method and Google's open-source human pose framework, Mediapipe BlazePose [2]. The muscle use and force/load parameter of RULA is calculated based on the VR work environment. Furthermore, we explore effective guidance methods through a user study by comparing visual, auditory, and combined auditory-visual guidance cues with an unguided condition.

We investigate the following research question: *How do real-time ergonomic posture adjustment guidance methods influence user behaviors while working in VR environments?* Our hypotheses in this study are:

- H_1 Providing guidance cues could improve users' ergonomic posture maintenance in VR.
- H_2 Combining audio and visual cues for ergonomic posture guidance could provide better posture adjustment than using audio and visual cues alone.

This work contributes to the design of real-time ergonomic posture guidance for VR working environments and presents a user study to explore effective posture adjustment interfaces.

2 RELATED WORK

There have been many studies on the use of machine learning in the prevention of work-related musculoskeletal disorders. Among these studies, a significant portion focuses on the development of intervention techniques, while research evaluating intervention methods and the implementation of effective interventions is relatively scarce [3]. Ding et al. [4] proposed a novel visual-based

method for real-time assessment of upper body posture for individuals sitting at desks while studying or using computers. Paudel and Choi [12] introduced an automated worker posture estimation system that computed workers' body angles and identified which angles were safe and unsafe while performing tasks in different workplaces. Kim et al. [8] presented a system based on OpenPose to calculate joint angles and RULA/REBA [6] scores. Most current research only automatically provides the corresponding risk scores. However, little research has focused on providing real-time adjustment methods in VR working environments. In this paper, we explore effective posture adjustment methods for VR working environments.

Audio and visual cues have been used to guide human posture adjustment. For example, Wu et al. [16] presented a method for whole-body pose modeling and quantitative assessment to identify and evaluate yoga poses, guiding learners. They employed wearable devices with 11 inertial measurement units (IMUs) fixed on the body to measure yoga pose data. Using the differences in angles between non-standard segments and a standard model, they guided in easily understandable language. Vignais et al. [14] introduced a system for real-time ergonomic assessment of manual tasks in industrial environments. They implemented computerized RULA ergonomic assessment using IMUs and provided visual feedback through a perspective head-mounted display. However, this visual guidance was separate from the environment and not integrated into the person's body. Additional visual cues and auditory warnings were given when predefined thresholds were exceeded. However, these studies required participants to wear additional body sensors, and the guidance was not intuitive and accurate enough.

In our system, we directly overlay visual guidance cues onto the user's body to provide intuitive visual feedback, presented simultaneously within the VR space. We also combine audio cues with video streams to provide guidance and feedback through multi-modal cues. In the next section, we describe our system in more detail and an experiment assessing the different guidance cues provided by the system.

3 PRELIMINARY TEST: MEDIAPIPE BLAZEPOSE PRECISION ASSESSMENT

There are a range of different methods for obtaining information about human body posture. Optical motion capture has high precision, but the equipment is complex and expensive. IMUs have drift errors and are burdensome to the human body. Kinect depth-sensing devices are not widely adopted and have occlusion problems. In contrast, machine vision-based human pose estimation only needs a webcam, offers a high level of ubiquity and affordability, and has no burden on the human body.

In our system, a lightweight human pose estimation method, Mediapipe BlazePose [2], was adopted to acquire human posture information. Due to the lack of precision in the human joint angles measured by the Mediapipe, a preliminary test was conducted to assess its precision and determine if it meets the system requirements.

Table 1: Marker place

Body Part	Location
Shoulder	Middle of humeral head laterally
Elbow	Lateral epicondyle of the elbow
Hip	Femoral greater trochanter
Wrist	Over lateral triquetrial bone
Knee	Lateral epicondyle of the femur
Nose	Tip of nose
Ear	Auditory meatus
Finger	5th metacarpal

3.1 Assessment Method

System and Experiment Design We conducted a preliminary experiment to assess the difference in the human pose data between Mediapipe and the ground truth. The ground truth for body movement was measured using the OptiTrack¹ motion capture system, which has exceptional precision. The experimental system was built with six OptiTrack cameras (Flex 13), a PC (Intel Core i7-8700 3.2GHz CPU with 6 Cores), and a Logitech C920 camera.

The experimental design is shown in Fig.1. Participants performed a set of identical movements with markers placed on their bodies while OptiTrack and Mediapipe simultaneously recorded the human posture data. The precision of Mediapipe human pose estimation was determined by analyzing the collected data and calculating the difference between OptiTrack and Mediapipe systems.

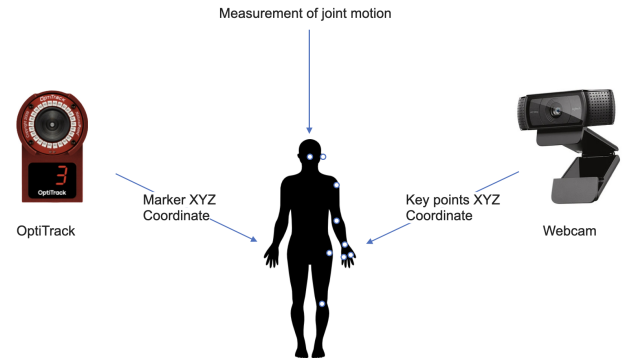


Figure 1: Mediapipe BlazePose Precision Assessment Experiment Design

The placement of markers on the participant was based on guidelines from the book *Measurement of Joint Motion: a Guide to Goniometry*[11], OptiTrack's official website for motion capture markers², Mediapipe's human body key points[2] and points required for RULA movements[9]. The specific positions of the markers on the human body are shown in the Table1.

Participants We recruited 14 participants from The University of Auckland, among whom two participants' data were excluded due to marker movement. The remaining 12 participants consisted of 9 males and 3 females, with an average age of 27.25 (SD = 5.987),

¹<https://optitrack.com>

²<https://docs.optitrack.com/markersets/full-body/biomech-57>

average height of 168.33 cm (SD = 9.18), and average weight of 64.25 kg (SD = 12.12). The participants had no musculoskeletal disorders or surgical history and could perform the required postures and motions.

Experimental Task The experimental task followed a within-group design, requiring each participant to sit on a chair and slowly perform a series of actions guided by the experimenters. These actions were determined based on the body parts and movements measured by RULA, as shown in Fig.2. There were five actions in total, each repeated three times. The order of the actions was from left to right, and the duration was around 5 minutes. The OptiTrack system automatically recorded the participant's human pose data. For Mediapipe BlazePose, a video of the human pose was first recorded by a Logitech C920 camera. Subsequently, the videos were input into the model for processing, extracting the coordinates of each frame's body key points and then output to a CSV file.

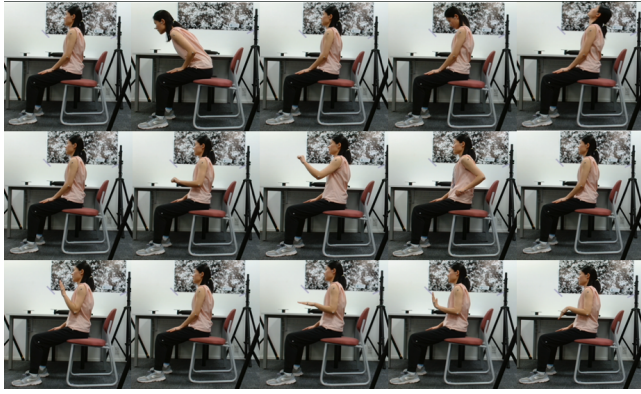


Figure 2: Participant's Movement in the Experiment

Experimental Procedure Before the experiment, it was necessary to calibrate the OptiTrack system daily. After the participants arrived, they were required to fill out a consent form and a basic information questionnaire. Before attaching reflective markers to the participant, they must enter the experimental environment and remove all reflective items from their bodies. Subsequently, the participants learned the required movements by watching instructional videos. After confirming that the participants could follow the instructions to perform the actions, eight reflective markers were attached to each participant. Once the participants were in the start position and indicated readiness, the formal experiment began. During the experiment, both the Mediapipe and OptiTrack systems were started, and the participants followed the instructions the experimenters gave. After completing the actions, both systems were stopped, and the participants were informed that the experiment was concluded, allowing them to relax. Next, the OptiTrack data was checked for any missing breakpoints caused by occlusion. If none were found, the experiment stopped; otherwise, the actions needed to be repeated. The entire experiment took approximately 60 minutes.

Data Processing Since the data for Mediapipe was captured at 30 fps and the data for Optitrack was 120 fps, it was necessary

Table 2: Point to Calculate Joint Angles

Body Part	Relevant point
Upper arm	elbow-shoulder-hip
Lower arm	wrist-elbow-shoulder
Wrist	pinky-wrist-elbow
Neck	ear-shoulder-hip
Trunk	shoulder-hip-knee

to downsample the OptiTrack data to 30 fps first. Subsequently, signal alignment was achieved through the alignsignal³ function in Matlab. After aligning the signals, joint angles for each frame were calculated using Matlab. The 3D coordinates of the three points were known, and the length of each side of the triangle formed by these points was calculated using Equation.1. Following this, the angles of the triangle were computed using Equation.2. The calculation method for each joint is illustrated in Table.2. After calculating the angles for each frame, the data could be compared. Equation.3 was employed to compute the absolute error of the i -th frame. Subsequently, Equation.4 and Equation.5 were used to calculate the mean and standard deviation (SD) of the absolute errors.

$$l = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} \quad (1)$$

$$\alpha = \cos^{-1} \left(\frac{a^2 + b^2 - c^2}{2ab} \right) \quad (2)$$

$$\alpha_i = |\alpha_o - \alpha_m| \quad (3)$$

$$\bar{\alpha} = \frac{(\alpha_1 + \alpha_2 + \dots + \alpha_i)}{i} \quad (4)$$

$$SD = \sqrt{\frac{\sum_{i=1}^n (\alpha_i - \bar{\alpha})^2}{n - 1}} \quad (5)$$

3.2 Assessment Result and Discussion

The model coordinates output by Mediapipe BlazePose come in two forms: one in real-world three-dimensional coordinates measured in meters, referred to as "world landmarks," and the other normalized coordinates for the length and width of the input video image, known as "pose landmarks." In this experiment, the human body's neck, trunk, upper arm, lower arm, and wrist joint angles were derived from both the Mediapipe world landmark model and the pose landmark model. The participant was photographed from a straight-side view. The 3D joint angles of the two models, the 2D joint angles without the depth information, and the joint angles calculated using the pose landmark model's input image pixel coordinates were compared with the ground truth data to obtain the system precision. The results are presented in the Fig.3.

It can be observed that the three-dimensional results from Mediapipe generally do not perform as well as the two-dimensional results, with the pixel-based calculation in the two-dimensional results being the most accurate. This is attributed to the fact that Mediapipe is a neural network model that takes two-dimensional

³<https://au.mathworks.com/help/signal/ref/alignsignals.html>

images as input, and its depth information is estimated, leading to lower three-dimensional accuracy. Through the experimental data, it can be concluded that the accuracy of calculating human joint angles using the two-dimensional pixel coordinates from Mediapipe is the highest and is within ten degrees of the RULA classification. Therefore, it meets the requirements of our system.

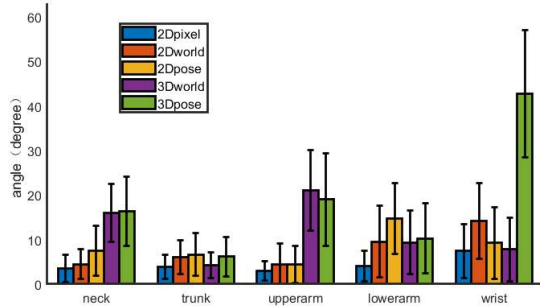


Figure 3: The Result of Mediapipe BlazePose Precision Assessment. Blue represents the precision of the pose landmark pixel joint angle, orange represents the precision of the 2D-world landmark joint angle, yellow represents the precision of the 2D-pose landmark joint angle, purple represents the precision of the 3D world landmark joint angle, and the green represents the precision of 3D pose landmark joint angle.

4 SYSTEM AND USER STUDY

4.1 System Design

Our ergonomic posture guidance system was mainly based on human pose estimation. Specifically, we deployed a standard webcam on the user's side for posture detection. When a person's posture deviates from the established standards, visual cues are overlaid on the real-time video stream captured by the camera. The back-end software performing the analysis was built based on RULA and Mediapipe. Augmented guiding cues on the webcam frame can be streamed into the VR workspace, reminding users to correct their posture in time. Additionally, when the RULA score surpasses a certain threshold, optional auditory cues (beep) are triggered. These visual and audible cues persist until the individual correct their posture, at which point they disappear.

To simulate the office environment in VR, we designed a testing scenario using the Unity game engine⁴. As shown in Fig. 4, a 24-inch monitor was placed 0.7 meters away from the participant in the virtual scene, displaying a 2,000-character Chinese text (text screen), and a small screen half the length and width of the screen was placed directly underneath it, displaying a real-time video stream and visual postural cues of the user (visual cue screen). We wanted people to focus more on their work (main task), so the big screen was in the near-peripheral eye, and the small screen was in the mid-peripheral eye. The user's priority in the workspace was to complete their work. For the hardware, we used the HTC Vive Pro

Eye head-mounted display⁵ connecting to a PC with an NVIDIA GeForce RTX 3060Ti graphics card and an Intel i5 processor. An Aoni webcam A20 was used for posture monitoring.

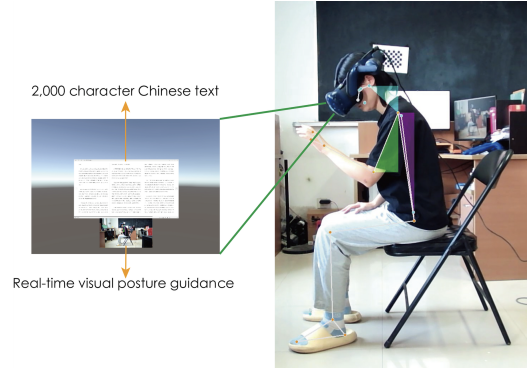


Figure 4: Experimental setup. The right picture shows a user in a VR headset with visual cues overlaid on the body. The left picture demonstrates the VR scenario consisting of a big text screen and a small visual cue screen.

4.2 Experimental Design

We designed a user study to compare three proposed ergonomic posture guidance methods (1)audio, (2) visual, and (3) audio-visual combined conditions, with a baseline condition (no guidance). In the baseline condition, the participants could only see the text and live video stream. The second condition involved reading the text along with auditory posture cues. The third condition included reading the text with visual posture cues. In the fourth condition, participants read the text while receiving both auditory and visual posture cues.

Participants We recruited 16 participants from Northwestern Polytechnical University, among whom two participants' data were excluded due to not triggering any posture guidance feedback. The remaining 14 participants consisted of 7 males and 7 females, with an average age of 25.71 (SD = 4.01), average height of 166.86 cm (SD = 8.42), and average weight of 59.82 kg (SD = 11.25). Only one participant had no prior experience with VR. All participants had normal or corrected-to-normal vision and were right-handed.

Experimental Task The experiment used a within-subject design where each participant was asked to read 2000 Chinese characters presented on the large screen described in subsection 4.1 under the four different posture guidance conditions. We used Latin Square to vary the order of conditions and balance the learning effect. Below the big screen, a smaller screen displayed a video stream of the participant on which the adjustment guidance was overlaid. During the task execution, users completed the reading task while adjusting their posture based on visual or auditory posture guidance cues. The system automatically recorded the time spent looking at each screen and calculated the RULA scores in real-time.

Experimental Procedure Before the experiment, each participant was given 5 minutes to read the instructions. Then, the experimenter helped the participants put on the VR headset and guided

⁴<https://unity.com/releases/editor/archive>

⁵<https://www.vive.com/hk/product/vive-pro-eye/overview/>

them through an eye-tracking calibration procedure. After the calibration, the experimenter ran a training VR program and asked the participants to familiarize themselves with the reading environment and posture guidance cues. Once the participants were familiar with the VR environment, the experimenter started the main reading task program. After each session, participants were asked to complete the System Usability Scale (SUS) [1], Simulator Sickness Questionnaire (SSQ) [15], and a 5-point Likert scale about subjective reports (see subsection 5.2). The whole experiment took around 60 minutes.

5 RESULTS

5.1 Performance

Gaze Duration on Text Screen We collected participants' gaze duration on the text through eye-tracking. The data contained outliers, and the distribution of some data points deviated from the normal (none: $p = .000$, audio: $p = .001$, vision: $p = .033$, audio-vision: $p = .000$). A Friedman test was run to determine if there were differences in average reading time across the four conditions. There were no statistically significant differences for the baseline (Mdn = 170.50s), audio (Mdn = 186.50s), vision (Mdn = 198.00s), and audio-vision conditions (Mdn = 179.00s), $\chi^2(2) = 3.686$, $p = .297$.

Gaze Duration on Visual Cue Screen The duration of the participants' gazes on the visual cue screen was also collected. There were some outliers in the data, and not all the data were normally distributed (none: $p = .955$, audio: $p = .001$, vision: $p = .374$, audio-vision: $p = .064$). A Friedman test was run, and pairwise comparisons were performed using a Bonferroni correction for multiple comparisons. Visual cue gaze duration was statistically significantly different between the conditions, $\chi^2(2) = 23.277$, $p < .0005$. Post hoc analyses revealed statistically significant differences in visual cue screen gaze time from baseline (Mdn = 6.00s) to audio-vision (Mdn = 12.50s) ($p = .016$) and vision (Mdn = 21.00s) ($p < .0005$), from audio (Mdn = 7.50s) to vision (Mdn = 21.00s) ($p = .010$), but not others.

Mean RULA Score We calculated the participants' RULA scores every single frame during the experiment based on the method presented in [10]. These RULA scores were recorded and stored to disk for further analysis. The Mean RULA score refers to the mean value of recorded RULA scores for each participant in each condition. The data did not contain outliers, but not all of them followed a normal distribution (none: $p = .008$, audio: $p = .135$, vision: $p = .002$, audio-vision: $p = .586$). Based on the Friedman test, there was a statistically significant difference in mean RULA score between conditions, $\chi^2(2) = 20.314$, $p < .0005$. Post hoc analyses showed a statistically significant decrease in mean RULA score from baseline (Mdn = 3.37) to audio-vision (Mdn = 2.06) ($p < .0005$), audio (Mdn = 2.09) ($p = .002$) and vision (Mdn = 2.12) ($p = .032$). However, no significance was found between the other conditions.

Percentage of Time with RULA Scores Greater Than or Equal to 3 RULA score can be categorized into different ranges [10]. For example, 1-2 means an acceptable posture, and 3-4 means further investigation and change may be needed. We filtered the frames with RULA scores greater than or equal to 3 and calculated the number of these frames. Then, we divided this number by the total frame number for each trial. We used this index to reflect the proportion of posture change requirements throughout each

trial. In our experiment, the highest score was 4. The dataset had no outliers, yet not all of them exhibited a normal distribution. (none: $p = .006$, audio: $p = .576$, vision: $p = .002$, audio-vision: $p = .398$). Based on the Friedman test, there was a statistically significant difference between conditions in the percentage of time with RULA scores > 3 (Fig. 5), $\chi^2(2) = 21.171$, $p < .0005$. Post hoc analyses showed a statistically significant decrease in the percentage of time with RULA scores > 3 from baseline (Mdn = 80.25%) to audio-vision (Mdn = 6.23%) ($p < .0005$), audio (Mdn = 8.55%) ($p = .005$) and vision (Mdn = 11.45%) ($p = .032$). However, no significant differences were observed in the remaining cases.

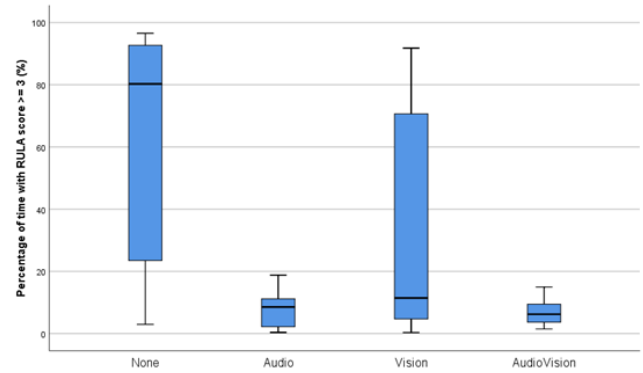


Figure 5: Percentage of Time with RULA scores bigger than or equal to 3.

5.2 Usability

System Usability Scale (SUS) The usability of all the systems was assessed by the System Usability Scale (SUS) questionnaire. Due to the presence of an outlier in the dataset, a Friedman test was conducted. The results indicated a statistically significant difference in usability across the different conditions, $\chi^2(2) = 24.956$, $p < .0005$. Post hoc analyses showed a statistically significant increase in usability from baseline (Mdn = 50.00) to audio-vision (Mdn = 85.00) ($p < .0005$) and audio (Mdn = 77.50) ($p = .002$), from vision (Mdn = 72.50) to audio-vision (Mdn = 85.00) ($p = .041$), while others did not exhibit significant differences.

Simulator Sickness Questionnaire (SSQ) We used the SSQ survey to assess simulator sickness. Friedman test was conducted because of outliers in the dataset. Statistical significance was not observed between any of the conditions with baseline (Mdn = 20.00), audio (Mdn = 19.00), vision (Mdn = 19.50), and audio-vision (Mdn = 19.50), $\chi^2(2) = 5.656$, $p = .130$.

Subjective Report We designed a 5-point Likert scale questionnaire⁶ based on [14] to understand the user experience. There were 9 dimensions: Understandability, sufficiency, ergonomic support, attention, ease of execution, intuitiveness and accuracy, efficiency, confidence, and enjoyment. As shown in Figure 6, audio-vision was the best in all the nine aspects. Vision performed better in efficiency, intuitiveness, accuracy, and sufficiency compared to audio, but it

⁶https://docs.google.com/document/d/1rZeJ3KU4LkHvIsoXfl8a7nKKPft6bQMW/edit?usp=share_link&ouid=107635347179654064260&rtfpof=true&sd=true

lagged behind in terms of confidence, enjoyment, ease of execution, and attention. Both methods showed similar performance in terms of understandability and ergonomic support.

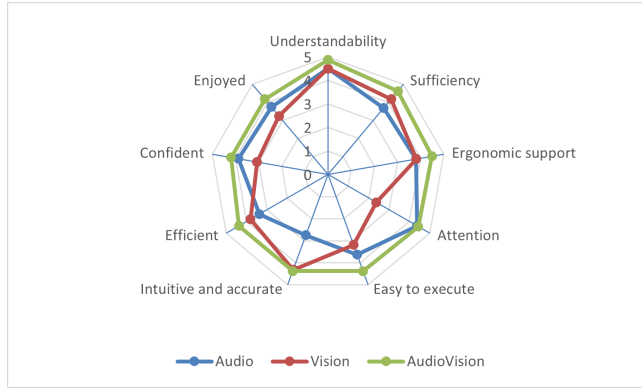


Figure 6: Radar chart of subjective reports (1: strongly disagree; 5: strongly agree; The more covered area, the better).

5.3 Discussion

Our results revealed that providing augmented ergonomic posture guidance in a VR workspace significantly reduced the time individuals spent in uncomfortable postures and their associated RULA risk scores, compared to no guidance cues. Thus our hypothesis H1 is supported. These methods did not have a significant impact on the overall task completion time, although the presence of visual cues increased the time participants spent gazing at the visual cue screen. Among all three ergonomic posture guidance methods, the combined auditory-visual method was the most effective method with the highest SUS score. Thus our hypothesis H2 was supported. The auditory method came next, followed by the visual method. This could be because when only visual cues were available, people might focus on the main tasks and ignore the visual cues. In contrast, the auditory-only cues could capture immediate attention but might not provide sufficient visually intuitive information. Some participants in the experiment expressed that the ergonomic posture guidance system could correct bad posture in time, and they believed our guidance interfaces could be useful to prevent musculoskeletal diseases caused by long-term bad working postures, especially in the immersive virtual environment.

However, some participants indicated that auditory cues occasionally caused distractions during their work. A softer sound or replacing auditory cues with tactile vibrations might be preferable. The existing system also requires placing a camera on the user's side, which could be inconvenient and raise privacy concerns for both the user and others. Further consideration is required for practical usage.

5.4 Conclusion and Future Work

In this work, we provided ergonomic posture adjustment guidance for people working in a VR environment and conducted a user study to compare three guidance methods. We found that the ergonomic posture guidance system could reduce people's time in

poor postures during VR office work, lowered posture risk scores, and had no significant impact on work duration. We also found that auditory cues could rapidly capture people's attention, while visual cues provided more comprehensive and effective posture guidance.

In the future, we plan to investigate the impact of using other cues like tactile feedback, aiming at exploring a more comfortable and natural user experience. We will also consider replacing the standard camera with other infrared or depth sensors to mitigate privacy concerns. Finally, we could use a 360-degree fish-eye camera near people's bodies to reduce inconveniences from distant camera placement.

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